

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	N. Parnifa	Reg. No.:	192525322
Section / Batch:	2025-2029	Date:	20.7.26

Instructions to Students

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type		Questions	Marks
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1 Q2,Q3	40 60 (2X30)
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

- Q1. Application : Smart Grid Energy Distribution System.
- Q2. Application : Natural Language Processing (NLP) Engine.
- Q3. Application: Video Compression System

Code of Conduct

I N. Parvika (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N. Parvika

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (40 Marks)						
Q2 (30 Marks)						
Q3 (30 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1) Application : Smart Grid Energy Distribution System

The given recurrence relation is:

$$T(n) = 4T(n/2) + n \log n$$

The standard form of the Master Theorem is:

$$T(n) = aT(n/b) + f(n)$$

Step 1: Identify the Parameters

Comparing the given recurrence with standard form:

$a = 4$ (number of subproblems)

$b = 2$ (size of each subproblem)

$$f(n) = n \log n$$

Step 2: Calculate $n \log_b a$

$$n \log_2 4 = n^2$$

Now compare:

$$f(n) = n \log n$$

$$n \log_b a = n^2$$

Since , $n \log n = o(n^{2-\epsilon})$

where $\epsilon > 0$, $f(n)$ grows slower than n^2 .

Step 3: Select the correct case

According to the Master Theorem, if

$$f(n) = O(n^{\log_b a - \epsilon})$$

then Case 1 applies.

Therefore,

$$T(n) = O(n^{\log_b a}) = O(n^2)$$

Interpretation Using Asymptotic Notation

Big - O : $O(n^2)$

Big - Ω : $\Omega(n^2)$

Big - Θ : $\Theta(n^2)$

Here, the final time complexity is

$$\boxed{\Theta(n^2)}$$

Conclusion:

The given recurrence satisfies Case 1 of the Master Theorem. Therefore, the Smart Grid Energy Distribution algorithm has time complexity of $\Theta(n^2)$.

2) Application: Natural Language Processing (NLP) Engine:

An NLP Engine processes text data using the recurrence relation.

$$T(n) = T(n/2) + n$$

Use the substitution method to solve the recurrence.

The recurrence relation is

$$T(n) = T(n/2) + n$$

Step 1: Expand the Recurrence

$$T(n) = T(n/2) + n$$

Substitute $T(n/2)$:

$$= T(n/4) + \frac{n}{2} + n$$

Again $T(n/4)$:

$$= T(n/8) + \frac{n}{4} + \frac{n}{2} + n$$

Continuing the expansion,

$$= T(n/2^k) + n \left(1 + \frac{1}{2} + \frac{1}{4} + \dots \right)$$

Step 2: Find the value of k

The recursion stops when

$$\frac{n}{2^k} = 1$$

$$2^k = n$$

$$k = \log_2 n$$

Step 3: Evaluate the series

$$1 + \frac{1}{2} + \frac{1}{4} + \dots$$

Therefore,

$$T(n) = \Theta(n)$$

Interpretation Using Asymptotic Notation

Big - O : $O(n)$

Big - Ω : $\Omega(n)$

Big - Θ : $\Theta(n)$

Hence,

$$\boxed{T(n) = \Theta(n)}$$

Conclusion:

Thus, the NLP Engine has linear time complexity, making it efficient and scalable for processing large text corpora.

3) Application : Video Compression System :

Question: Analyze all three using asymptotic Notation.

The three algorithms have the following time complexities:

⊛ Algorithm A: $O(n^{1.5})$

⊛ Algorithm B: $O(n \log n)$

⊛ Algorithm C: $O(n^2)$

Numerical Comparison

Let the input size be $n = 1000$

Algorithm A: $n^{1.5} = 1000^{1.5} = 31623$

Algorithm B: $n \log_2 n = 1000 \times 10 \approx 10000$

Algorithm C: $n^2 = 1000^2 = 1000000$

Comparison Table

Algorithm	Complexity	Algorithmic Operations
A	$O(n^{1.5})$	31,623
B	$O(n \log n)$	10,000
C	$O(n^2)$	1,000,000

Growth Order

$$O(n \log n) < O(n^{1.5}) < O(n^2)$$

This shows that $O(n \log n)$ is the most efficient among the three algorithms.

Interpretation Using Asymptotic Algorithms:

Best : $O(n \log n)$

Moderate : $O(n^{1.5})$

Least Efficient : $O(n^2)$

Conclusion:

Among the three algorithms, $O(n \log n)$ is the best choice for video compression because it has the lowest growth rate and provides the fastest execution for large video files.



	$O(n \log n)$	A
	$O(n^{1.5})$	B
	$O(n^2)$	C